

WORK EXPERIENCE

Space Exploration Technologies Corp. (SpaceX)		Hawthorne, CA, USA
Raptor Combustion Simulation · Distributed GPU-accelerated fluid dynamics simulations of combusting flows · C++/CUDA		
Propulsion Software Engineer II	Details forthcoming	2025-12 – Present
Propulsion Software Engineer	Details forthcoming	2025-01 – 2025-12
Software Engineering Intern		2023-08 – 2023-12
• 25% speed-up of the Raptor Combustion CFD Code via CUDA C++ GPU kernel optimizations.		

AWARDS

• Gordon Bell Prize Finalist · Association for Computing Machinery (ACM)	2025-01
200+ trillion grid points on 43,000 GPUs · Conference Paper Press: ORNL · LLNL · HPCwire · Georgia Tech	
• Verkadathon Winner · “Best Use of Verkada Products” · Verkada · Company hackathon · as an intern	2023-07

PUBLICATIONS

FULL: [GOOGLE SCHOLAR](#)

• Simulating many-engine spacecraft: Exceeding 1 quadrillion degrees of freedom [...].	Paper · 2025
• MFC 5.0: An exascale many-physics flow solver.	Paper · 2025
• Open-Source Combusting Flow Simulation.	Undergraduate Thesis · 2024

UNDERGRADUATE RESEARCH

Computational Physics Group (CPG) · Multi-Component Flow Code (MFC)	2021-11 – 2024-12
• Implemented reactive combustion flow modeling and case-specific compile-time optimizations.	
• Accelerated the solver via OpenACC on OLCF Summit & Frontier; added CMake, integration testing, and CI.	
Space Systems Design Laboratory (SSDL) · GT-2 Satellite	2021-11 – 2022-12
• Implemented the satellite’s over-the-air update protocols, on bare-metal, from ground to orbit.	

EDUCATION

Georgia Institute of Technology (GT) · Bachelor’s in Computer Science · Diploma	2021 – 2024
GPA: 3.82 · Research Option · Modeling & Simulation / Systems & Architecture	Atlanta, GA, USA

MISCELLANEOUS

• Student Volunteer at Supercomputing (SC) · Denver, CO and Atlanta, GA	2023-11 & 2024-11
• Languages: English (Native, United States Citizen) and French (Native, French Citizen)	